Experiment 8
Properties of Radiation

SAFETY

Make sure that you have read the **General Safety Notes**, in the Introductory section of this manual, before you begin.

Do not, **under any circumstances,** attempt to repair any of the equipment. If you suspect equipment to be faulty, turn it off at the power point and talk to your demonstrator.

It is a general rule that you **must not eat or drink** in the laboratories. This is **extremely important** when dealing with radioactive materials. The substances you will be handling today are not especially dangerous under normal conditions – but if they enter your body they can be **extremely harmful**, even fatal. If you are thirsty, use the water fountain at the eastern end of the laboratories.

To minimise your exposure to radiation, always be sure to:

- Keep the radioactive source as far away from you as possible.
- Minimise the time that you are exposed to the source. Always use the shiny Aluminium plate to shield from the electrons.
- Shield the radioactive source (using the lead blocks provided) whenever possible.
- Wash your hands after handling the source and at the end of the lab session.

The lead shielding blocks are also toxic. Wash your hands after handling them to prevent lead poisoning.

The radiation detectors used in the laboratory are powered by a **high voltage** power supply. If any leads or components appear damaged, talk to your demonstrator immediately.

Outline of Laboratory Exercise

In **Part A** you will examine either the effects of increasing distance from a radioactive source or the quantitative effects of radiation shielding. The insights you will gain will hopefully help you to understand the effectiveness of radiation protection strategies - the ways in which a combination of distance and shielding can reduce radiation exposure.

In **Part B** you will design your own investigation by examining one other aspect of the properties of radiation.

✔***Pre-Lab Exercise*****:** *Read the entire laboratory exercise. You must pay particular attention to the relevant Safety Notes for this exercise, as well as the discussion of Radioactivity in the Introduction (below). Complete the online pre-lab tasks.*

RADIOACTIVE MATERIALS

In this exercise you will be dealing with samples of radioactive material. When you receive a source of radioactivity from your demonstrator, **you must sign for it** on the radioactive sources record. When you return the radioactive source to your demonstrator at the end, **you must sign AGAIN** on the radioactive sources record.

You MUST return all radioactive materials and sign the record for this before you leave.

Learning Goals

- To use the method of statistics to analyse a large data set including determining the statistical uncertainty of the data.
- To explore exponential rates of change graphically and use this to determine the relationship of the variables.
- To understand and explore how radiation distribution and strength changes with distance.
- To develop an understanding of the nature of radiation and the effect of shielding.

Introduction – Radioactivity

The term **radioactivity** refers to any process in which atoms of a particular element spontaneously change into atoms of a different element. Sometimes this happens when an unstable nucleus ejects part of itself as an **alpha** particle (which is actually two protons and two neutrons bonded together – i.e. a Helium nucleus). Alpha particles are the most damaging form of radiation, but they do not penetrate shielding easily – they will usually be stopped by any material as dense as a piece of paper.

Alpha decay is relatively uncommon. More often, elemental change happens because the **nucleons** (neutrons and protons) inside the nucleus change – either a proton turns into a neutron, or vice versa. This type of change is accompanied by the emission of a charged particle (called a **beta** particle) and/or other radiation from the atom's nucleus. Beta particles and gamma rays are generally less damaging but more penetrative than alpha particles – they require more dense shielding materials.

(A beta particle is actually either an electron or an anti-electron. Anti-electrons are often called positrons.)

When a nucleus **decays**, the 'daughter' nucleus (i.e. the nucleus after it has changed) is often in an excited state. This is analogous to atomic excitation, when electrons are raised into unstable higher-energy states – but when a nucleus is excited, it is not electrons but instead the nucleons which are in higher-energy states. To release the excess energy, the protons and neutrons emit **gamma rays**: high-energy photons of electromagnetic radiation. Gamma rays from nucleons are usually at much higher energies (~MeV) than photons from electrons (e.g. X-rays) – the energy difference between an excited state and a ground state is much larger for nucleons than for electrons.

The radioactive source you will be using in today's exercise is a sample of Cobalt-60. This transforms via beta decay (β^-) into an excited nucleus of Nickel-60, which then emits a gamma ray (**γ**) with an energy of 1.2 MeV as it de-excites. In nuclear physics terms, this reaction is represented as follows:

$$^{60}Co \rightarrow {}^{60}Ni^* + \beta^- + \overline{\nu}$$

$$^{60}Ni^* \rightarrow {}^{60}Ni \quad + \gamma$$

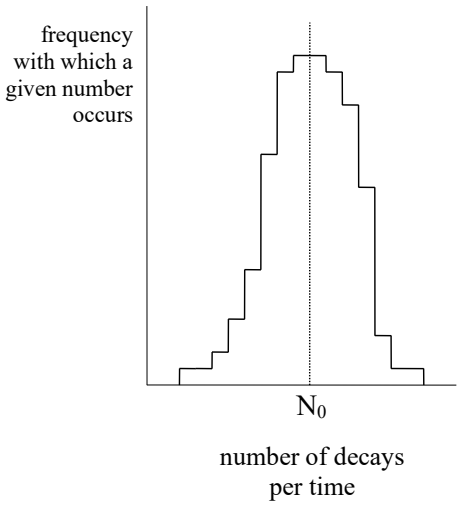
Your source is covered by a thin lead sheet, which will absorb everything but the gamma rays. This means that your source should appear to be producing 1.2 MeV gamma rays only.

The Randomness of Radioactive Decay

Radioactive materials decay one nucleus at a time. Each nucleus decays at random, and independently of the other atoms around it. Because of this randomness, if we count the number of decays over a certain length of time and then repeat our measurement, we don't expect to get the exact same number of decays each time. However, the **probability** that a nucleus will decay is not completely random – it is determined by the structure of the nucleus. Since all the nuclei have the same structure, every nucleus has the same probability of decay.

The variation to expect in the number of decays over a given time interval can be predicted using the statistics of probability. If we count the number of decays per minute over and over again, statistical analysis leads us to expect that the different numbers measured will cluster around a central average value $\mathbf{N_0}$ (the **mean**):

This mean value $\mathbf{N_0}$ is the number that best represents the average rate of decay. But if we make only one measurement, how close to $\mathbf{N_0}$ do we expect our measurement to be? Statistical analysis shows that for a normal distribution of randomness, 68.3% of measurements will be within ± one **standard deviation** of the mean, 95.4% of measurements will be within ± two standard deviations, and 99.97% will be within ± three standard deviations.



For the kind of randomness exhibited by radioactive decay (which is called a Poisson distribution), it turns out that the standard deviation is equal to the square root of the mean: $\sqrt{N_0}$. http://hyperphysics.phy-astr.gsu.edu/hbase/Math/poifcn.html

Because we know that radioactive decay follows this statistical pattern, we can use this standard deviation value as our statistical ± **uncertainty** value (see also Appendix A). So if we make a single measurement which gives us a number **N**, then we can be 68.3% sure that the actual mean value is within the range:

$$N \pm \sqrt{N}$$

We cannot know the mean from a single measurement, but this statistical analysis lets us know how close our measurement must be to the mean, within a definite range of uncertainty. This is useful.

To take account of the background radiation, we must find **N** as a result of the total number of counts $\mathbf{N_t}$ minus our number of background counts $\mathbf{N_b}$. Instead of taking the square root of **N**, we must then take the square root of our total measurements. Considering background radiation, our result is:

$$(N_t - N_b) \pm \sqrt{N_t + N_b}$$

Note that although we are subtracting the two measurements (total and background) to get **N**, we can't subtract our two uncertainties – it doesn't make sense for our total uncertainty to get smaller! Therefore, when calculating uncertainties we add them together instead.

Statistical Uncertainty — Example

Suppose you measure a total number of counts (i.e. from the radioactive source in addition to background radiation) of N_t = 1200. You also measure background counts for the same length of time as N_b = 400.

Therefore: $N = N_t - N_b$ = 1200 – 400 = 800

And the uncertainty is: $\sqrt{(N_t + N_b)} = \sqrt{(1200 + 400)} = \sqrt{1600} = 40$

So our final data result is written: N = 800 ± 40

This means that we have a statistical 68.3% probability that the 'actual' mean value is a number between 760 and 840. Note that this uncertainty has nothing to do with uncertainties in our apparatus, or any kind of other experimental error – the uncertainty here is entirely due to the **statistical randomness** of radioactive decay.

REMEMBER: you **must** calculate statistical uncertainty values for all your measurements.

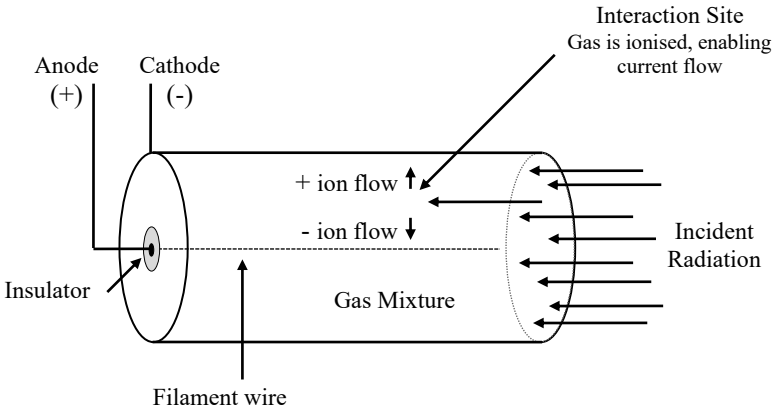
The inverse square law describes how the intensity of radiation falls off with distance, which is important to consider when dealing with radioactive sources.

(read more here: http://hyperphysics.phy-astr.gsu.edu/hbase/forces/isq.html).

The expected relationship between number of counts and distance is $N_x = \frac{k}{x^2}$

Geiger Counters

In the following laboratory exercise you will be using a **Geiger counter** to detect radiation. A Geiger counter consists of a cylindrical case containing a mixture of gases, with a wire running along its axis. The wire is insulated from the case and held at a positive electrical potential (voltage) relative to the case, using a DC power supply.



When radiation enters the counter, it ionises some of the gas molecules in its path. The negative ions or free electrons created by this ionisation are attracted to the counter's anode (via the wire), while positive ions are attracted to the counter's cathode (via the case). They are accelerated by the electric field difference between cathode and anode (due to the voltage from the power supply). As the charges accelerate they collide with other gas molecules, thereby causing **more** ionisation, and more free charges, and even more ionisation. This process of **charge multiplication** occurs many times. Ultimately, a substantial negative charge will be deposited on the anode. The current flow that results between cathode and anode is employed to produce the Geiger counter's 'clicking' sound.

In general: radiation entering the Geiger counter is used to produce electrical current, which provides the power to electronically detect (and 'click') that the radiation exists.

Background Radiation

You should notice that the radiation counter will detect continual radiation even when the radioactive source is not nearby. This is **background radiation** – the constant small amount of radiation that exists everywhere around us, all the time. Ordinary concrete and bricks contain traces of radioactive material (e.g. Thorium, Potassium), and we are also bombarded by many **cosmic rays** from outer space.

When measuring the radioactivity of a particular source, it is obviously necessary to first measure the average background radiation, and then to subtract this background from your measured data.

Part A

Your demonstrator has a radiation source and in front of you is a geiger counter. Read the information in the following link for a quick overview on the workings of a geiger counter http://hyperphysics.phy-astr.gsu.edu/hbase/nuclear/geiger.html.

The radiation source is cobalt-60 which emits a 1.2 MeV photon (as well as an electron). The electrons will be absorbed by the thin Aluminium sheet(shiny) covering the source but let through the gamma rays.

Your demonstrator will show you how to use your Geiger counter apparatus.

SAFETY NOTE

Do **not** change the high voltage setting on your radiation detector.

Be sure to use **both hands** when adjusting the position of the Geiger tube.

In the laboratories so far, you have learnt how to experimentally validate physics concepts. In your groups, decide whether you will investigate how radioactivity varies with distance OR how radioactivity varies with shielding. Choose one of these physics concepts to validate for Part A.

Radioactivity and Distance

In your groups, decide how you will validate experimentally the inverse square law. Determine the experimental setup, how you will collect the data, how many data points you will collect and how many trials you will need. Discuss your ideas with your demonstrator.

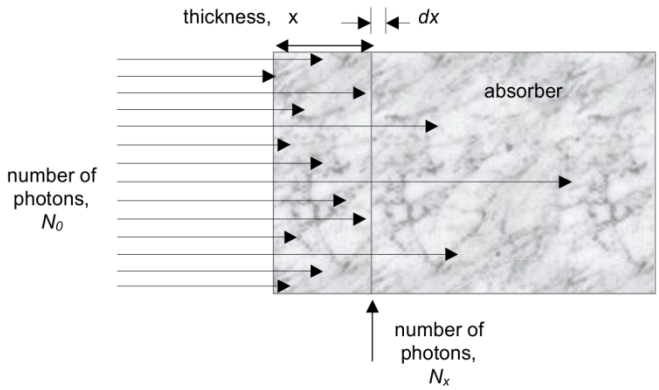
- Consider the following: how will you account for the background radiation? How will you measure the distances from source to detector?
- Write out the method in your notebook. This can be in dot points but should include a labelled diagram of the experimental setup.
- Collect the data and plot an appropriate graph. Hint: consider how you can linearise the data. Include uncertainties.

Analysis

Comment on your results. Is it expected? Why/why not? Your analysis should link to relevant physics concepts, and include an explanation of any discrepancies, a discussion of sources of errors and possible improvements to your experimental procedure.

Radioactivty and Shielding

When radiation passes through material the intensity is reduced. This reduction is also known as attenuation.



The amount of radiation absorbed is dependent on the thickness of the shielding material. The change in the radiation intensity is given by the following equation:

$$dN = -\mu N_x dx$$

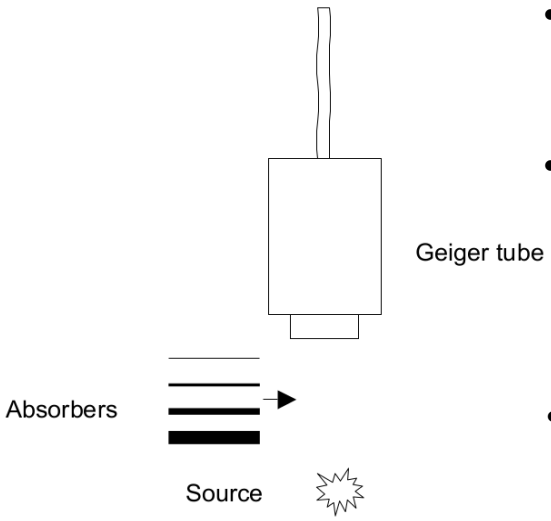
The μ is called the linear attenuation coefficient and will depend on the radiation type and the shielding material. Note this is different to the decay constant, l .

Integrating the equation above we find that the intensity of the radiation at a depth x into the shielding is given by

$$N_x = N_0 e^{-\mu x}$$

where N_0 is the amount of radiation hitting the surface of the shielding.

Now we can test the attenuation on the gamma rays created by lead shielding.



- You will have two types of shielding, Aluminum (shiny) that will stop beta particles and lead in different thicknesses.
- Once again, determine the experimental setup and discuss with with your demonstrator. Write a brief method in your notebook with a fully labelled diagram.

Remember the inverse square law so do not change the distance to the Geiger tube during these measurements.

- Collect the data and plot an appropriate graph in order to find the attenuation coefficient. Hint: consider how you can linearise the data. Include uncertainties.

Analysis

Comment on your results. Is it expected? Why/why not? Your analysis should link to relevant physics concepts, and include an explanation of any discrepancies, a discussion of sources of errors and possible improvements to your experimental procedure.

- Using some algebra on our equation for N_x, how would you suggest you find the value of μ? Calculate μ for the lead with gamma rays as the radiation.
- How thick would the shielding need to be to stop 90% of the radiation? How about 99%?

Part B

For Part B of this experiment, in your groups you can choose to investigate the other physics concept outlined above OR come up with your own investigation. Discuss your ideas with your demonstrator before you start.

Some options to get you thinking:

- Investigate how radioactivity varies with different absorbers
- Investigate how radioactivity varies with angle
- Investigate how to determine radiation type using a magnet